Method		Winogrande		CSQA		HellaSwag		SIQA		PIQA		Avg	
		ACC↑	TR↓	ACC↑	TR↓	ACC↑	TR↓	ACC↑	TR↓	ACC↑	TR↓	ACC↑	TR↓
Few-shot Answer		57.1	-	66.8	-	45.3	-	68.2	-	61.6	-	59.8	-
Chain-of-Thought		56.7	43.1	69.9	28.8	45.4	34.8	67.3	37.3	62.7	41.3	60.4	37.1
Self-Consistency		56.4	40.9	72.4	24.9	46.2	33.7	64.7	41.0	54.6	45.8	58.9	37.3
Self-Refine		48.7	50.3	55.5	47.0	42.0	37.2	65.1	38.8	50.8	48.3	52.4	44.3
Least-to-Most		58.9	38.8	69.0	27.5	31.3	39.4	66.3	36.6	65.7	32.2	58.2	34.9
Contrasive CoT		59.4	38.0	71.1	23.2	45.9	35.6	67.0	36.0	68.1	44.9	62.3	35.5
Ours	RD Only	58.6	29.5	72.2	13.5	49.4	24.0	69.4	20.8	66.3	26.2	63.0	22.8
	SPS Only	59.1	22.8	72.6	18.9	49.3	18.7	69.7	28.8	68.5	27.6	63.7	23.4
	RIDERS	**60.7**	**16.6**	**73.2**	**6.7**	**50.6**	**12.8**	**71.9**	**15.3**	**69.9**	**15.9**	**65.3**	**13.5**

Table 1: Performance comparison across five commonsense reasoning datasets on Llama2-13B.

$$
\begin{aligned}
d_1 &= \mu L_q + L_c + \lambda L_c \\
d_2 &= \lambda L_c + L_q + \mu L_q \\
\Delta d &= d_2 - d_1 = L_q - L_c
\end{aligned} \tag{7}
$$

where d_1 is the total distance in normal serial positions and d_2 is the distance after swapping the two components. In most scenarios, we have $L_q < L_c$, thus, we can infer that $\Delta d < 0$. That means, if we replace the original order of "*[Question] + [CoT]*" with the order of "*[CoT] + [Question]*", we can not only increase the intensify of information flow from the question to the final prediction, but also reduce the total information loss due to the reduction in total distance. Although this method is straightforward in implementation, it proves to be effective in both theory and experiments.

5.2 Mitigation Experimental Settings

Datasets Following previous works, we use five representative commonsense reasoning benchmarks: **WinoGrande** (Sakaguchi et al., 2020), **CSQA** (Talmor et al., 2019), **HellaSwag** (Zellers et al., 2019), **SIQA** (Sap et al., 2019) and **PIQA** (Bisk et al., 2020). The specific information of each dataset is reported in Appendix F.

Metrics In addition to the commonly used **Accuracy (ACC)** metric, we also introduce a new metric — **Toxic Rate (TR)**, to quantify the severity of Toxic CoT problems:

$$
TR(f) = |C_d \cap W_f| / |W_f| \tag{8}
$$

where C_d denotes questions that models give correct answers directly and W_f denotes questions that models give wrong answers after applying method f. Thus, we can infer that the lower the TR, the fewer Toxic CoT problems the method introduces.

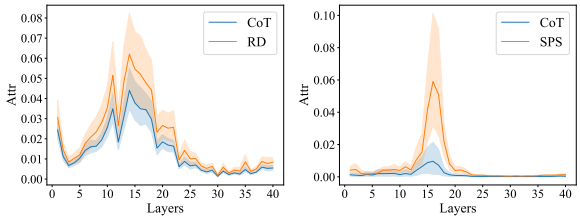
Baselines As our research focuses on enhancing CoT methods in commonsense reasoning, we select some of the latest CoT-like methods applicable to this task for comparison: **Few-shot Answer**, **Chain-of-Thought** (Wei et al., 2022), **Self-Consistency** (Wang et al., 2023c), **Self-Refine** (Madaan et al., 2023), **Least-to-Most** (Zhou et al., 2023) and **Contrasive CoT** (Chia et al., 2023). For all methods, we employ a 5-shot prompt and use 4 NVIDIA GeForce RTX 3090 GPUs for inference. More implementation details can be found in the Appendix F.

Method	Winogrande		CSQA	
	Type1	Type2	Type1	Type2
CoT	0.0	0.0	0.0	0.0
RD Only	41.9	47.8	56.7	46.2
SPS Only	58.1	78.3	46.7	**92.3**
RIDERS	**74.2**	**82.6**	**73.3**	84.6

Table 2: Accuracy on the two types of drifting issues.

5.3 Mitigation Results

The main result of our experiments on Llama2-13B is shown in Table 1. (The results on more models are presented in Appendix F.) We can get the following conclusions: **(1) Our method effectively mitigates Toxic CoT problems.** Compared to CoT prompting, our method reduces the Toxic Rate by an average of **23.6%** across five datasets. Besides, compared to other advanced CoT-like methods, our method causes the fewest Toxic CoT problems (decreased by an average of **22.0%** over SOTA methods). **(2) Our method can also improve the model's overall performance on commonsense reasoning.** Our work improves the accuracy on all benchmarks (improved by **5.5%** compared to CoT and **3.0%** compared to SOTA methods on average). This proves that the Toxic CoT problem poses a bottleneck in LLM's commonsense reasoning, highlighting the value of our work.



(a) RD method (b) SPS method

Figure 8: Information flow comparison on Winogrande after applying our two methods.

Method	ProofWriter		FOLIO	
	ACC↑	TR↓	ACC↑	TR↓
CoT	56.8	19.9	45.1	11.6
RIDERS	**78.2**	**6.4**	**48.0**	**7.5**

Table 3: Performance comparison across two logical reasoning tasks on Llama2-13B.

5.4 Discussion and Analysis

Performance on Two Drifting Issues To demonstrate the effectiveness of our method in addressing the Rationale Drift issue (Type1) and Answer Drift issue (Type2), we conduct experiments on these samples and report the results in Table 2. Both of our methods can mitigate the corresponding issues (RD solves **49.3%** Rationale Drift issue on average, while SPS resolves **85.3%** Answer Drift issue on average). This verifies the validity of our hypothesis 1, 2, as all of these methods are built upon them. Besides, combining the two methods leads to even greater improvements, demonstrating the necessity of optimizing from both of these perspectives.

Performance in the Model In §3 and §4, we probe information loss in two issues by tracing the information flow in models. Here, we repeat the attribution tracing experiments, comparing the differences before and after applying our method to further validate the effectiveness of our work. As we can see from Figure 8 and 16, our two methods (orange lines) increase the information flow from questions in two stages compared to CoT prompting (blue lines). This indicates our method indeed compensates for the information loss in the LLM.

Performance on Other Tasks Our work mainly focuses on commonsense reasoning tasks. To further evaluate the effectiveness of our approach on other types and forms of tasks, we conduct the main experiments on two logical reasoning tasks: **ProofWriter** (Tafjord et al., 2021) and **FOLIO** (Han et al., 2022)). As presented in Table 3, we can find that the Toxic CoT problem also exists in other tasks, and our method can still mitigate it.

Method	Wino	CSQA	Hella	SIQA	PIQA	Avg
CoT	3.2k	2.6k	4.2k	2.9k	2.7k	3.1k
SC	4.5k	3.2k	5.5k	3.8k	4.3k	4.3k
SR	9.2k	6.9k	10.2k	7.6k	6.5k	8.1k
L2M	8.7k	7.9k	11.4k	8.3k	6.7k	8.6k
CON	4.6k	3.5k	5.4k	3.8k	3.5k	4.2k
Ours	3.6k	2.9k	4.9k	3.4k	3.3k	3.6k

Table 4: Token consumption per example comparison.

Cost Analysis For the applicability, we measure the computation and time cost of our approach. Here we compare the token cost between our method and the baseline. According to Table 4, our method requires fewer tokens compared to other SOTA methods (only 1.2× cost of the basic CoT method). We also compare the time cost of our decoding method in Appendix G and find that the speed of RD is comparable to existing decoding strategies. Therefore, we illustrate the cost-efficiency of our approach across different tasks.

6 Related Work

6.1 CoT Problems Analysis and Mitigation

Recently, many works have focused on analyzing and mitigating problems in CoT reasoning. For analytical work, most studies focus on black-box LLMs. Through intervening or paraphrasing prompts and comparing outputs, researchers can interpret the reasons leading to errors in the model's reasoning (Lanham et al., 2023; Liu et al., 2023c; Wang et al., 2023a). For optimization works, they design additional supervision signals or training processes for the model (Ramnath et al., 2023; Liu et al., 2023a) or leverage external resources for the model (Shinn et al., 2023; He et al., 2023; Lyu et al., 2023). However, these works lack the probing of inner mechanisms behind these problems, leading to insufficient analysis or less universally applicable optimization methods.

6.2 Mechanistic Interpretability

The work on mechanistic interpretability aims to understand the internal mechanisms of models when performing various tasks. Early work focused on how the model stores factual knowledge internally (Meng et al., 2022; Dai et al., 2022). In recent times, some research efforts have shifted towards examining how models retrieve and utilize knowledge. This includes internal knowledge

retrieval (Geva et al., 2023), knowledge retrieval from prompts (Wang et al., 2023b; Jin et al., 2024b), and the utilization of knowledge for reasoning purposes, such as math reasoning (Stolfo et al., 2023) and multi-step reasoning (Hou et al., 2023). However, there is limited existing work that explains commonsense reasoning and CoT reasoning, which are significant contributions of this work.

7 Conclusion

In this paper, we find a problem named Toxic CoT, which results in the model's reasoning deviating from the original correct answer when utilizing CoT-like prompting. Through tracing the internal information flow of the LLM with attribution tracing and causal tracing methods, we prove that this problem is mainly caused by the model's lack of information from the question in shallow attention layers when generating rationales or answers. Based on this result, we propose the RIDERS method to mitigate the Toxic CoT problem from both decoding and serial position perspectives. Through extensive experiments on multiple commonsense reasoning datasets, we verify the effectiveness of our approach in mitigating Toxic CoT problems and enhancing the model's overall commonsense reasoning capabilities.

Limitations

Although our work conducts an in-depth interpretation and mitigation of the Toxic CoT problem, it has several limitations. Firstly, like former commonsense reasoning works (Liu et al., 2022, 2023a; Xie et al., 2023), our research focuses on the form of multi-choice questions. This stems from the absence of effective evaluation methods for open-ended commonsense reasoning, leading to the predominance of benchmarks in this format. This calls for advancements in benchmark-related research. Secondly, we refrain from analyzing Toxic CoT problems in more reasoning tasks such as math, primarily due to the poor performance of current moderately-sized white-box models on these tasks. For instance, Llama2-13B achieves a mere 7.2% accuracy on GSM8K (Cobbe et al., 2021) without utilizing the CoT technique. This calls for developments in model-related research. We leave these limitations as our future work to explore.

Acknowledgement

This work is supported by the Strategic Priority Research Program of Chinese Academy of Sciences (No. XDA27020203), the National Natural Science Foundation of China (No. 62176257,62276095).

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